

I am interested in understanding safety risks that arise when humans interact with sociotechnical systems, particularly in **low-resource** and **socio-culturally** diverse contexts. My research focuses on harms associated with these systems, ranging from **online abuse** to **algorithmic and AI-mediated risks**. I aim to employ large-scale, data-driven methods with human-centered studies to understand how such harms are implemented and experienced by individuals and communities. These interests have been shaped by my experience working in **civic tech**, developing **user-centric** tools to address online harms in resource-constrained contexts.

Through research in natural language processing and network science, I seek to analyze how harmful content emerges, spreads, and is moderated across online platforms, and how technical systems can be designed to better support safety, agency, and inclusion for the majority of the world. I want to expand on these themes by analyzing how platform policies, recommendation systems, and AI systems collectively shape harmful content dynamics and user experience, while designing **Human-AI** systems that prioritize user agency, cultural context, and safety outcomes.

Online Platform-Mediated Harms: For my undergraduate thesis, I studied the spread of toxic content on online social networks. While prior work has largely relied on activation-based and epidemic models [1], I conducted an empirical analysis across three platforms¹, Twitter, Koo, and Gab, and found that these approaches did not adequately capture key patterns in the propagation of hateful content. To address this, I developed and evaluated a network-based model that treats users who amplify, attenuate, or replicate toxic content. This work resulted in two first-authored publications [2,3] and helped me develop a deeper understanding of quantitative research design, modeling choices, and the iterative nature of research.

Building on this work, I became interested in other forms of platform-mediated harm. One such growing risk is the proliferation of **financial scams** on social media, in regions with ambiguous legal frameworks like India. This translates into financial and social harm to people, particularly from marginalized backgrounds [4]. I wanted to understand how such harms move **cross-platform** and how they are experienced by those affected. To investigate this, I worked with **Dr. Kiran Garimella** from **Rutgers University** on a **mixed-methods** study of the betting app ecosystem in India [5]. I analysed three complementary data sources: large-scale organic and paid promotional content from social media platforms, like Meta; qualitative and quantitative annotations of harms present in these promotions; and sentiment and thematic analysis of user reviews from the Google Play Store. The study examined 23.6K Meta advertisements, 2.5K Instagram posts, and 329K app reviews.

I developed a qualitative annotation guideline and performed annotations of Meta Ads. I identified harmful practices like misleading advertising, deceptive celebrity endorsements, and the use of illegal deepfakes to feign legitimacy. Analysis of user reviews revealed recurring themes of financial loss, addiction, and fraudulent payout practices. Through this work, I gained valuable experience in mix-methods research, academic writing and learnt how critical empirical evidence can inform platform governance and policy design. Platform-mediated harms remain unevenly distributed, particularly in areas with limited oversight. Understanding how these harms move across platforms, how they are experienced by users, and how technical interventions can mitigate them is essential for creating safer and more inclusive digital spaces.

Human-AI Interaction: Having studied online abuse, I became interested in how technical systems can be designed to address the risks that emerge when new technologies interact with diverse and often under-served communities. I worked as a **Research Engineer** at **Tattle**², a civic tech non-profit in India, where I built open-source tools and datasets to understand and respond to **inaccurate** and **harmful** content.

My early work at Tattle focused on creating a Twitter dataset³ on **online gender-based violence** using a community-centered **participatory** approach [6]. We worked with people at the end of online abuse and identified gaps in existing responses to harm. This informed the design of Uli, a browser plugin tool that redacts abusive content online in low-resource languages like Hindi and Tamil. This approach, grounding **technical design in lived experience**, was the core framework of building tools at Tattle. This informed my contributions to other tools like, Feluda⁴, which supports multilingual and multimodal content analysis for fact-checkers, and the Deepfakes Analysis Unit⁵, a platform for analyzing and responding to deepfakes and AI-generated media.

I also investigated safety risks introduced by the rapid deployment of large language models (LLMs), particularly as chatbot-based systems are increasingly adopted by civil society organizations in resource-constrained settings. I led a comparative evaluation of LLM performance across Indian languages and found clear performance disparities relative to English [7]. Building on this,

¹Dataset size: Twitter (100K users, 17.2M posts), Koo (215K users, 16.9M posts), and Gab (62K users, 20.1M posts) and 2.3M edges.

²<https://tattle.co.in/>

³This won an **Outstanding Paper** Award at the **Workshop of Online Harms and Abuse** (WOAH) at **NAACL'24**

⁴<https://tattle.co.in/products/feluda/>

⁵DAU, Media Coverage: [NY Times, The Nieman Lab]

I collaborated with MLCommons⁶ to develop an AI safety benchmark for Hindi, focused on hate speech and sex-related harms in the Indian context [8]. Benchmark prompts were co-created with Hindi-speaking domain experts who either identify as, or have extensive experience working with, communities affected by identity-based targeting online. I also examined user-facing risks of LLM chatbots deployed by NGOs in health and education settings in India, identified unsafe responses to harmful queries, and worked with these organizations to integrate safety guardrails using Llama-Guard models and context-specific slur lexicons.

More recently, I have become interested in understanding how people actually engage with AI systems in practice, particularly how they **seek** and **evaluate information** through AI versus traditional web search, and how these behaviors vary across demographic and cultural contexts. In ongoing work with Dr. Kiran Garimella, I am collecting and analyzing real-world human-AI interaction data from users in the Global South [9], examining how these exchanges unfold across diverse backgrounds and what they reveal about the conditions under which AI systems can be made safer and more socio-culturally aligned.

Future Research Interests: Drawing from my research experience, I am interested in working on research directions like,

(1) **Platform-Mediated Harm & Governance:** I want to look at how harms propagate across platforms, how governance on one platform affects adjacent platforms? What is the impact of social media recommendation systems on spread of harmful content/polarization? Can we design better algorithmic curation processes? I want to study how different content moderation interventions shape subsequent user behavior, whether they reduce harm, shift content patterns, or generate unintended effects on participation. Given that LLMs are decent at simulating human behavior, could we use them to conduct algorithmic audits?

(2) **Human-AI Interaction:** How do people from different socio-cultural backgrounds engage with AI systems in information-seeking contexts? How will LLMs reshape the online information ecosystem—in terms of user behavior, content dynamics, and trust? How do we design culturally and value-aligned AI systems for communities underrepresented in current AI development and evaluation? How do we measure and mitigate downstream harms in high-stakes domains such as health, education, and online safety?

Online platforms are sociotechnical systems shaped by the interplay of human behavior, algorithmic design, and policy. I hope my work can contribute to understanding these dynamics and inform better system design and governance, especially as technologies like LLMs grow in capability and scale.

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⁶This work was included in the [Alluminare](#) benchmark suite, assessing generative AI across 12 safety hazards.